

Bayesian Estimation of Informed Trading in Prediction Markets

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March 2025

Abstract

What fraction of trading on prediction markets is driven by private information versus noise? We estimate the Probability of Informed Trading (PIN) on 2,263 resolved binary Polymarket markets (~ 29 million trades, September 2024 – February 2025) using four Bayesian prior specifications: static independent (M1, conjugate Gibbs), hierarchical partial pooling (M2, MH-within-Gibbs), regime-switching (M3, FFBS), and time-varying (M4, single-site MH). MLE produces boundary solutions for δ in 84.4% of markets; the Bayesian approach produces zero. M2’s hierarchy reduces credible interval widths for 100% of markets, with the largest gains in thin markets. M3 reveals that 99.4% of markets alternate between quiet ($\text{PIN} \approx 0.11$) and active ($\text{PIN} \approx 0.55$) regimes; M4 finds similar dynamics but with critically low ESS (median 16). High-PIN markets are better calibrated (Brier scores 0.049–0.060 vs. 0.075–0.082 for low PIN), and M3’s PIN is the strongest predictor of accuracy ($\beta = 2.88$, $p = 0.013$), even though M1 wins on WAIC. Uninformed traders are systematically optimistic ($\varepsilon_b/\varepsilon_s = 1.65$) in a sample where 83% of events resolve NO; informed traders profit by correcting this bias, pushing prices toward truth. Unlike equity markets where PIN represents adverse selection cost, in prediction markets PIN signals forecast quality.

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1 Introduction

1.1 Prediction Markets as Probability Forecasts

Prediction markets like Polymarket are increasingly treated as probability forecasts. CNN displays Kalshi odds on-screen during elections, hedge funds feed Polymarket prices into quantitative models, and forecasters such as Nate Silver compare their predictions against prediction market prices. All of this assumes that a contract trading at \$0.72 means there is approximately a 72% chance the event occurs.

The theoretical foundation comes from [Wolfers and Zitzewitz \(2006\)](#), who show that under log utility and symmetric beliefs, prediction market prices exactly equal the mean belief across traders. Whether price equals probability depends on assumptions about trader preferences and belief distributions that we cannot observe directly. Empirically, [Reichenbach and Walther \(2025\)](#) provide the most comprehensive assessment to date: using 124 million trades from Polymarket, they find that market prices closely track realized probabilities, with no evidence of a general longshot bias. However, they identify biases including a persistent tendency to overtrade the “Yes” option and mispricing concentrated in the early and late stages of a market’s lifecycle.

[Clinton and Huang \(2025\)](#) provide a contrasting perspective. Analyzing nearly 2,500 political prediction markets across PredictIt, Kalshi, Polymarket, and the Iowa Electronic Markets for the 2024 U.S. presidential election, they find substantial variation in accuracy across platforms — and Polymarket, despite having the highest trading volume, was the least accurate at 67%. These findings raise a natural question: what explains the variation in accuracy across markets? One possibility is that the information structure differs — some markets may have more informed traders pushing prices toward truth, while others may be dominated by noise or expressive trading.

1.2 Informed Traders and the PIN Model

The PIN model is only useful if informed trading is a real phenomenon in these markets. [Bosschaerts et al. \(2024\)](#) provide direct evidence from field prediction markets: they identify “price-sensitive” traders whose trades systematically push prices toward the true outcome, consistent with [Kyle’s \(1985\)](#) theoretical framework. These traders comprise roughly 7% of participants but are responsible for most of the informational content in prices.

This motivates PIN estimation on Polymarket. The Probability of Informed Trading (PIN), originally developed by [Easley et al. \(1996\)](#) for equity microstructure, decomposes observed buy and sell order flows into informed and uninformed components. If informed traders exist and drive price accuracy, then PIN should be a meaningful quantity. Higher PIN could mean *better* calibration (informed traders push prices toward truth) or *worse* calibration (adverse selection distorts prices). We do not assume a direction; discovering the sign is an empirical finding.

1.3 The MLE Problem and the Bayesian Alternative

The PIN model has been estimated by maximum likelihood since Easley et al. (1996). Griffin et al. (2021) demonstrate that MLE is systematically problematic. For five large-cap stocks over twelve years, MLE produces corner solutions — the news-direction parameter δ estimated at exactly 0 or 1 — on 34–50% of trading days. The Bayesian estimates produce zero corner solutions. The average size of daily PIN changes is 25–48% smaller under the Bayesian method, and the variance of these changes is 37–68% lower.

Griffin et al.’s Bayesian approach uses a fully conjugate Gibbs sampler with data augmentation, which we adopt as our baseline. However, they estimate each stock independently with a single static model. Our project extends their approach in four directions: hierarchical pooling across markets within the same category (M2), regime-switching dynamics allowing the information environment to shift between discrete states (M3), time-varying dynamics allowing informed trading intensity to drift continuously (M4), and systematic prior sensitivity analysis.

Because prediction market events resolve, we observe $\omega \in \{0, 1\}$ and can test whether estimated PIN relates to price accuracy — a validation opportunity unavailable in equity markets where PIN is typically estimated.

2 Data

2.1 Source and Sample

Trade-level data from Polymarket, obtained via the prediction market analysis dataset.¹ Our sample consists of 2,263 resolved binary markets spanning September 2024 through February 2025, comprising approximately 29 million trades. This period covers the 2024 U.S. presidential election (Trump vs. Harris), Federal Reserve rate decisions (50bp cut in September through a pause in January), cryptocurrency events (Bitcoin surpassing \$100K), and sports markets (NFL playoffs, Super Bowl LIX). Markets are categorized into four groups: Politics, Sports, Crypto & Finance, and Other.²

Selection criteria require: (1) ≥ 20 trading days; (2) ≥ 10 average daily trades (after zero-filling non-trading days); and (3) $\geq 25\%$ two-sided trading days (days with both $B > 0$ and $S > 0$). Markets that did not clearly resolve to YES or NO were excluded. The final 24 hours of trading before resolution are also excluded, because in those hours the outcome is effectively known and nearly all trades are informed, which would artificially inflate PIN.

2.2 Order Flow Definition and Processing

Raw trade data — consisting of timestamp, maker/taker identity, side, size, price, and market ID — is aggregated to daily buy and sell counts: (B_{jd}, S_{jd}) for each market j on day d . These two

¹<https://github.com/Jon-Becker/prediction-market-analysis>

²Categorization was performed iteratively using LLM-assisted classification of market question text, with manual review and correction until all markets were assigned to appropriate categories.

integers per market-day are the only inputs to the PIN model. In our primary specification, B_{jd} counts buy-YES plus sell-NO orders, and S_{jd} counts sell-YES plus buy-NO orders. This definition follows from how the DGP defines δ as the probability of good news: a buy-YES and a sell-NO both express the belief that the event will occur, so they belong on the same side of the order flow. The daily data also includes closing prices, volume-weighted average prices, and USDC trading volume, which are used in the calibration analysis but not in PIN estimation.

Market metadata includes the resolution outcome $\omega_j \in \{0, 1\}$ (did the event occur?), category, lifespan, total volume, and price statistics (mean, median, first, and final prices). Crucially, ω_j is *not* used in PIN estimation — it is reserved for validation, where we use it to assess whether markets with higher estimated PIN produce prices that more accurately predict ω_j (Section 6).

2.3 Descriptive Statistics

We began with approximately 20,900 markets from Polymarket spanning March 2023 through November 2025. A first pass requiring ≥ 10 trading days and ≥ 50 trades retained roughly 6,600 markets. A second pass applied stricter quality filters: ≥ 20 trading days, ≥ 10 average daily trades (after zero-filling non-trading days), and $\geq 25\%$ two-sided trading days (days with both $B > 0$ and $S > 0$). We then restricted the time window to September 2024 through February 2025 due to computational constraints, focusing on a period with high market activity and diverse event types. After excluding markets without clear binary resolution and removing the final 24 hours before resolution, the final analysis sample contains 2,263 markets.

Table 1: Descriptive statistics by category.

Category	Markets	Mean Days	Median Days	Avg Trades	% YES
Politics	862	55.0	57.0	14,290	22.5%
Sports	634	99.0	108.5	21,061	8.5%
Crypto & Finance	451	47.5	32.0	5,184	17.5%
Other	316	43.5	33.0	3,355	14.9%
Total	2,263	64.2	57.0	—	17.0%

The overall median duration is 57 days (mean 64.2, range 20–173). Sports markets are the longest-lived (median 108.5 days), while Crypto & Finance and Other markets are shortest (medians 32 and 33 days). Across all categories, only 17% of markets resolve YES — meaning 83% resolve NO. This asymmetry becomes important in Section 6, where we show that uninformed traders are systematically optimistic. Of the 2,263 markets, 24.9% have fewer than 30 trading days, making them particularly susceptible to the MLE boundary problems discussed in Section 4.1.

3 The PIN Model

3.1 Intuition

The PIN model classifies each trading day into one of three latent types: (1) *no information event* — only noise traders are active, buys and sells are roughly balanced; (2) *good news* — someone knows the event is likely to happen, so there is an extra burst of buying YES and selling NO (both counted as “buys” B_{jd} in our order flow definition) on top of the usual noise; (3) *bad news* — someone knows the event is unlikely, so there is an extra burst of selling YES and buying NO (both counted as “sells” S_{jd}). The model’s job is to infer, from observed buy and sell counts alone, which days had information events and how intense the informed trading was.

3.2 Likelihood

The model (Easley et al., 1996) posits a daily generative process governed by five parameters per market j :

α_j	probability that an information event occurs on day d
δ_j	probability that news is good, given an information event
μ_j	expected informed trades per event day
$\varepsilon_{b,j}$	expected uninformed buys per day
$\varepsilon_{s,j}$	expected uninformed sells per day

For each trading day: (1) an information event occurs with probability α_j ; (2) if an event occurs, it is good news with probability δ_j ; (3) uninformed buys and sells arrive as $\text{Poisson}(\varepsilon_{b,j})$ and $\text{Poisson}(\varepsilon_{s,j})$; (4) on good-news days, additional informed buys arrive as $\text{Poisson}(\mu_j)$; on bad-news days, additional informed sells arrive as $\text{Poisson}(\mu_j)$.

This gives a three-component Poisson mixture over latent day types $z_{jd} \in \{1, 2, 3\}$. The day-level likelihood is:

$$\begin{aligned}
 L_{jd} = & \underbrace{(1 - \alpha_j) \cdot f(B_{jd} \mid \varepsilon_{b,j}) \cdot f(S_{jd} \mid \varepsilon_{s,j})}_{\text{no event: noise only}} \\
 & + \underbrace{\alpha_j \delta_j \cdot f(B_{jd} \mid \varepsilon_{b,j} + \mu_j) \cdot f(S_{jd} \mid \varepsilon_{s,j})}_{\text{good news: } \mu_j \text{ extra buys}} \\
 & + \underbrace{\alpha_j (1 - \delta_j) \cdot f(B_{jd} \mid \varepsilon_{b,j}) \cdot f(S_{jd} \mid \varepsilon_{s,j} + \mu_j)}_{\text{bad news: } \mu_j \text{ extra sells}}
 \end{aligned} \tag{1}$$

where $f(\cdot \mid \lambda)$ is the Poisson probability mass function. The likelihood sums over the three possibilities because the day type is unobserved. The first term says: with probability $1 - \alpha_j$, no information arrived today, so buys and sells occur at the baseline noise rates. The second term says: with probability $\alpha_j \delta_j$, good news happened, so buys now arrive at rate $\varepsilon_{b,j} + \mu_j$ — the usual noise plus additional informed buying — while sells remain at $\varepsilon_{s,j}$ because informed traders only trade in the

direction of the information they receive. The third term handles bad news symmetrically. The full market likelihood is $L_j = \prod_{d=1}^{T_j} L_{jd}$. This likelihood is shared by all four specifications; they differ only in their priors.

3.3 PIN as a Derived Quantity

The PIN itself is:

$$\text{PIN}_j = \frac{\alpha_j \mu_j}{\alpha_j \mu_j + \varepsilon_{b,j} + \varepsilon_{s,j}} \quad (2)$$

The numerator $\alpha_j \mu_j$ is the expected number of informed trades per day (probability of an event times the arrival rate of informed traders given an event), and the denominator adds the uninformed trades. PIN is the fraction of all trading that comes from informed participants.

PIN is a *derived* quantity — it has no prior and no likelihood of its own. It is a nonlinear function of five parameters, so we compute it from every MCMC draw rather than plugging in posterior means. The nonlinearity means the posterior mean of PIN $\neq$ PIN evaluated at the posterior means of the components (Jensen’s inequality). The joint posterior also exhibits a ridge structure: high α with low μ produces similar PIN to low α with high μ , which creates challenges for MCMC sampling.

3.4 Data Augmentation and Full Conditionals

The mixture likelihood (Equation 1) sums over unobserved day types, which makes it difficult to isolate individual parameters for posterior sampling. Following Griffin et al. (2021), we use data augmentation to decouple them. At each MCMC iteration, two augmentation steps are applied:

Step 1: Day-type imputation. For each day, we compute how probable each latent type is given the current parameters and observed counts:

$$P(z_{jd} = 2 \mid B_{jd}, S_{jd}, \theta_j^{(s)}) = \frac{\alpha_j^{(s)} \delta_j^{(s)} \cdot f(B_{jd} \mid \varepsilon_{b,j}^{(s)} + \mu_j^{(s)}) \cdot f(S_{jd} \mid \varepsilon_{s,j}^{(s)})}{L_{jd}^{(s)}}$$

and then draw z_{jd} from these probabilities. Once day types are known, α_j ’s full conditional reduces to counting event vs. non-event days (Beta kernel), and δ_j reduces to counting good-news vs. bad-news days (Beta kernel).

Step 2: Trade split. On event days, informed and uninformed trades arrive from two independent Poisson processes, so the number of informed trades given the total is a Binomial draw. On good-news days, informed buys are drawn as $B_{jd}^i \sim \text{Binomial}(B_{jd}, \mu_j/(\mu_j + \varepsilon_{b,j}))$; on bad-news days, informed sells are drawn as $S_{jd}^i \sim \text{Binomial}(S_{jd}, \mu_j/(\mu_j + \varepsilon_{s,j}))$. This separates informed from uninformed trades, giving μ_j a Gamma full conditional (counting informed trades) and $\varepsilon_{b,j}, \varepsilon_{s,j}$ Gamma full conditionals (counting uninformed trades).

These two steps are identical across all four specifications. What differs is whether the priors are conjugate to the resulting likelihood kernels — yielding closed-form full conditionals and pure

Gibbs sampling (M1, M3) — or whether non-conjugate priors require Metropolis–Hastings steps (M2, M4). Table 2 summarizes the full conditionals and sampling methods across all specifications.

Table 2: Full conditionals and sampling methods across all specifications. Top panel: PIN model parameters (enter the Poisson likelihood via data augmentation). Bottom panel: specification-specific parameters (updated given latent states, do not enter the Poisson likelihood directly).

Spec.	Parameter	Prior	Kernel	Posterior	Sampler
<i>PIN model parameters (enter likelihood):</i>					
M1	α_j	Beta(1, 1)	$\alpha^{n_{\text{ev}}}(1-\alpha)^{n_{\text{no}}}$	Beta($1+n_{\text{ev}}$, $1+n_{\text{no}}$)	Gibbs
	δ_j	Beta(1, 1)	$\delta^{n_{\text{good}}}(1-\delta)^{n_{\text{bad}}}$	Beta($1+n_{\text{good}}$, $1+n_{\text{bad}}$)	Gibbs
	μ_j	Ga(1, .01)	$\mu^{\Sigma_{\text{inf}}}e^{-n_{\text{ev}}\mu}$	Ga($1+\Sigma_{\text{inf}}$, $.01+n_{\text{ev}}$)	Gibbs
	$\varepsilon_{b,j}, \varepsilon_{s,j}$	Ga(1, .01)	$\varepsilon^{\Sigma_{\text{uninf}}}e^{-T\varepsilon}$	Ga($1+\Sigma_{\text{uninf}}$, $.01+T$)	Gibbs
M2	θ_j (all 5)	LogN(m_k, s_k^2)	same as M1	LogN $\times$ Beta/Gamma	MH
M3	$\alpha_{j,r}$	Beta(1, 1)	$\alpha_r^{n_{\text{ev}}^{(r)}}(1-\alpha_r)^{n_{\text{no}}^{(r)}}$	Beta($1+n_{\text{ev}}^{(r)}$, $1+n_{\text{no}}^{(r)}$)	Gibbs
M4	$\alpha_j(t)$	$N\left(\frac{\ell_{t-1}+\ell_{t+1}}{2}, \frac{\sigma_\eta^2}{2}\right)$	$\alpha_t^{\mathbb{K}[\text{ev}]}(1-\alpha_t)^{\mathbb{K}[\text{no}]}$	LogitN $\times$ Beta	MH
<i>Specification-specific parameters (do not enter likelihood):</i>					
M2	$m_{\theta,k}$	$N(0, 10)$	$\prod_{j \in k} N(\log \theta_j \mid m_k, s_k^2)$	$N(\bar{\ell}_k, s_k^2/n_k)$	Gibbs
	$s_{\theta,k}^2$	Inv-Ga	$\prod_{j \in k} N(\log \theta_j \mid m_k, s_k^2)$	Inv-Gamma	Gibbs
M3	p_{qq}, p_{aa}	Beta(9, 1)	$p^{n_{\text{stay}}}(1-p)^{n_{\text{switch}}}$	Beta($9+n_{\text{stay}}$, $1+n_{\text{switch}}$)	Gibbs
M4	σ_η^2	Inv-Ga(2, 1)	$(\sigma_\eta^2)^{-T/2}e^{-\sum \eta_i^2/2\sigma_\eta^2}$	Inv-Ga($2+\frac{T}{2}$, $1+\frac{\sum \eta_i^2}{2}$)	Gibbs

[†]M3 and M4: $\delta_j, \mu_j, \varepsilon_{b,j}, \varepsilon_{s,j}$ same as M1 (Gibbs).

[‡]M4: $\ell_t = \text{logit}(\alpha_t)$; prior is Normal on ℓ_t conditional on neighbors; kernel is Beta from one day only.

Latent sequences are also sampled each iteration: M3’s regime sequence $\{s_d\}$ via FFBS, and M4’s $\alpha(t)$ path via single-site MH. These are not parameters but latent states.

4 Four Prior Specifications

All four specifications share the same likelihood (Equation 1). They differ in how α behaves across days — and in M2’s case, how all parameters relate across markets. The specifications encode four competing hypotheses about how the information environment works in prediction markets.

The default priors for the five PIN parameters, used by M1 and shared as the base by M3 and M4, are deliberately vague:

Parameter	Support	Prior	Rationale
α_j	[0, 1]	Beta(1, 1)	Flat: equal weight on all event rates
δ_j	[0, 1]	Beta(1, 1)	Flat: equal weight on news direction
μ_j	[0, ∞)	Gamma(1, 0.01)	Mean = 100, SD = 100: allows 1–10K
$\varepsilon_{b,j}$	[0, ∞)	Gamma(1, 0.01)	Same: wide enough for any market
$\varepsilon_{s,j}$	[0, ∞)	Gamma(1, 0.01)	Same

After 20–30 days of (B, S) count data, the likelihood overwhelms these priors — posteriors are driven almost entirely by observed data.

4.1 M1: Static Independent PIN (Conjugate Gibbs)

Structural assumption. The information environment is constant over the life of each market, and each market is unrelated to every other market. Five parameters per market, estimated from that market’s data alone.

MCMC implementation. Because M1 uses conjugate priors throughout, the data augmentation described in Section 3.4 yields a pure Gibbs sampler with closed-form full conditionals:

$$\alpha_j \mid z \sim \text{Beta}(1 + n_{\text{event}}, 1 + n_{\text{no}}) \quad (3)$$

$$\delta_j \mid z \sim \text{Beta}(1 + n_{\text{good}}, 1 + n_{\text{bad}}) \quad (4)$$

$$\mu_j \mid z, B^i, S^i \sim \text{Gamma}\left(1 + \sum B_{jd}^i + \sum S_{jd}^i, 0.01 + n_{\text{event}}\right) \quad (5)$$

$$\varepsilon_{b,j} \mid z, B^i \sim \text{Gamma}\left(1 + \sum (B_{jd} - B_{jd}^i), 0.01 + T_j\right) \quad (6)$$

and similarly for $\varepsilon_{s,j}$. There are no Metropolis–Hastings steps and no tuning parameters.

4.2 M2: Hierarchical Partial Pooling (MH-within-Gibbs)

Structural assumption. Markets within the same category share a common information environment. Individual market parameters are drawn from a category-level distribution, so a thin political market can borrow strength from data-rich political markets. All five parameters receive hierarchical priors:

$$\log(\theta_j) \sim N(m_{\theta,k}, s_{\theta,k}^2) \quad \text{for category } k,$$

where $\theta \in \{\alpha, \delta, \mu, \varepsilon_b, \varepsilon_s\}$. Category means receive weakly informative priors $m_{\theta,k} \sim N(0, 10)$.

Prior sensitivity. The category-level variance $s_{\theta,k}^2$ controls how much borrowing occurs. We fit M2 under three hyperprior specifications:

Label	Specification	Effect
Tight	$s_{\theta,k}^2 \sim \text{Inv-Gamma}(10, 1)$	Concentrates markets near category mean
Moderate	$s_{\theta,k}^2 \sim \text{Inv-Gamma}(2, 1)$	Allows moderate heterogeneity
Diffuse	$s_{\theta,k}^2 \sim \text{Inv-Gamma}(0.5, 0.5)$	Barely constrains heterogeneity

MCMC implementation. The hierarchical prior on the log scale breaks the conjugacy that made M1 fully Gibbs. Specifically, α_j now has a Beta-kernel likelihood (from the day-type augmentation) times a LogNormal prior (from the hierarchy) — this product is not a known distribution, so α_j requires a Metropolis–Hastings step. The same applies to $\mu_j, \varepsilon_{b,j}, \varepsilon_{s,j}$. Category-level means

$m_{\theta,k}$ have conjugate Normal full conditionals (Normal–Normal), and variances $s_{\theta,k}^2$ have conjugate Inverse-Gamma full conditionals. The day-type and trade-split augmentation steps (Section 3.4) are shared across all specifications.

4.3 M3: Regime-Switching PIN (FFBS)

Structural assumption. The information environment switches between two discrete states. In a “quiet” regime ($r = 1$), few information events occur (low α); in an “active” regime ($r = 2$), informed traders are present and α is high. Each regime gets its own event probability $\alpha_{j,r} \sim \text{Beta}(1, 1)$. The market transitions between regimes according to a Markov chain with transition matrix:

$$\mathbf{P} = \begin{pmatrix} p_{qq} & 1 - p_{qq} \\ 1 - p_{aa} & p_{aa} \end{pmatrix}$$

where p_{qq} is the probability of staying in the quiet regime and p_{aa} is the probability of staying active. Priors $p_{qq}, p_{aa} \sim \text{Beta}(9, 1)$ encode the belief that regimes are sticky (expected persistence ≈ 0.9).

MCMC implementation. The regime structure fundamentally changes the sampling. In M1, each day’s type z_{jd} is drawn independently. Here, the regime state $s_{jd} \in \{1, 2\}$ depends on $s_{j,d-1}$ through $\mathbf{P}$, so the entire regime sequence must be sampled jointly via the forward–filtering backward–sampling (FFBS) algorithm. The forward pass computes filtered probabilities $P(s_{jd} \mid B_{j,1:d}, S_{j,1:d})$ recursively; the backward pass draws the full sequence $\{s_{j,1}, \dots, s_{j,T}\}$ conditional on the filtered probabilities. Each regime’s $\alpha_{j,r}$ is then updated via Gibbs using only the days assigned to that regime, and the transition probabilities update conjugately:

$$p_{qq} \mid \{s_d\} \sim \text{Beta}(9 + n_{\text{stay quiet}}, 1 + n_{\text{switch to active}})$$

and similarly for p_{aa} . The remaining parameters $(\delta_j, \mu_j, \varepsilon_{b,j}, \varepsilon_{s,j})$ are updated as in M1.

4.4 M4: Time-Varying PIN (State-Space, Single-Site MH)

Structural assumption. Information intensity evolves continuously:

$$\text{logit}(\alpha_j(t)) = \text{logit}(\alpha_j(t-1)) + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2) \quad (7)$$

with $\sigma_\eta^2 \sim \text{Inv-Gamma}(2, 1)$. The drift variance σ_η^2 controls how much α can change from one day to the next. When $\sigma_\eta^2 \rightarrow 0$, the daily shocks η_t vanish and $\alpha_j(t)$ is effectively constant across all days, collapsing M4 to M1. When σ_η^2 is large, $\alpha_j(t)$ can jump dramatically between consecutive days, producing behavior that resembles M3’s discrete regime switches. In this sense, M4 nests both M1 and M3 as limiting cases, with σ_η^2 governing where along that spectrum the data places the model.

MCMC implementation. Each day’s $\alpha_j(t)$ is sampled via single-site Metropolis–Hastings: the prior on $\text{logit}(\alpha_t)$ is Normal conditional on its neighbors, but the Poisson likelihood is not conjugate to this Gaussian state evolution, requiring an MH accept/reject at each time step. This means a market with T days requires T correlated MH updates per iteration, creating a high-dimensional sampling problem.

5 Results

Table 3 summarizes the MCMC settings, convergence, and total runtimes across all four specifications.

Table 3: MCMC settings, convergence diagnostics, and total runtimes across specifications.

Model	Iterations	Burn-in	Thin	Saved	Med. ESS	Min ESS	Runtime
M1	12,000	4,000	4	2,000	2,000	25	~12h
M2	15,000	5,000	5	2,000	1,575	20	~3h [†]
M3	12,000	4,000	4	2,000	1,878	7	~65h
M4	15,000	5,000	5	2,000	16	2	~112h

[†]M2: ~1h per hyperprior specification $\times$ 3 specifications. M1, M3, M4 parallelized across independent market-level chains; M2 runs as a single joint chain. Runtimes are total CPU time across all cores, not wall time.

All models were implemented in R, with M2’s likelihood evaluation written in Rcpp/RcppArmadillo (~40% speedup). Estimation was run on the University of Chicago Midway3 HPC cluster (SLURM, caslake partition). Test runs on subsets of 3–5 markets were used for debugging and tuning on a local machine before scaling up.

5.1 M1: Static Independent PIN

Because M1 is a fully conjugate Gibbs sampler, mixing is excellent: median ESS across all markets is 2,000 (i.e., every saved draw is effectively independent), with a minimum of 25. Figure 1 shows traceplots for a representative market; all four parameters and the derived PIN show rapid mixing with no visible trends or stuck periods.

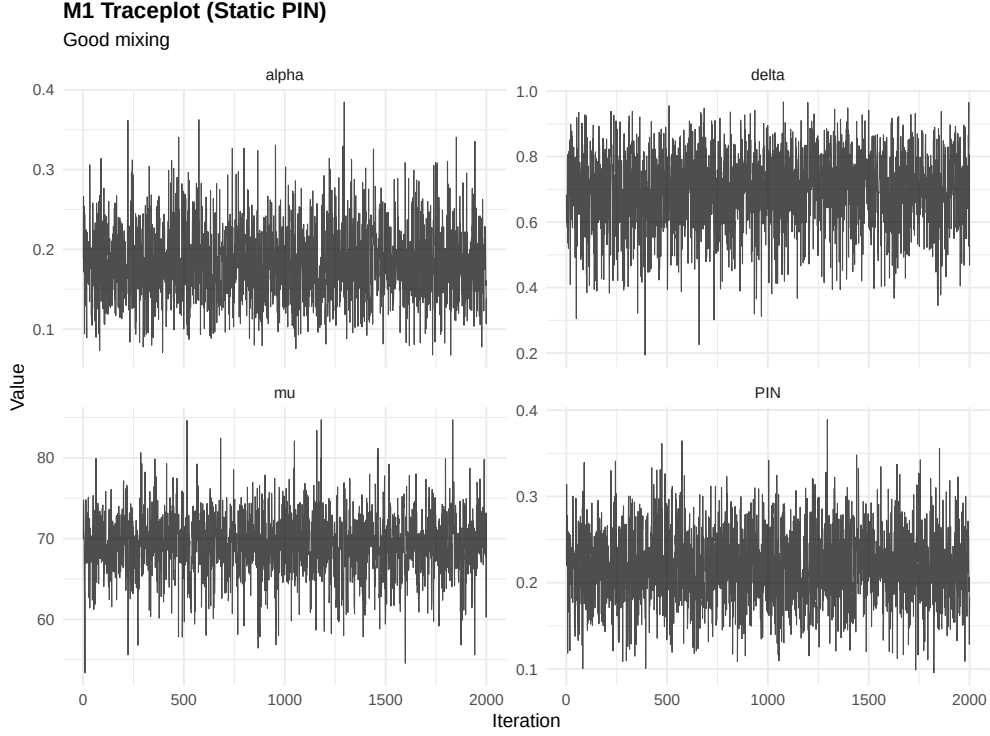


Figure 1: M1 traceplots for a representative market (ESS = 2,000). All parameters mix rapidly, as expected from the fully conjugate Gibbs sampler.

M1 produces median PIN of 0.292 across all 2,263 markets with a median 95% credible interval width of 0.202. Table 4 summarizes the posterior distributions across all markets.

Table 4: M1 posterior summary across 2,263 markets.

Parameter / Quantity	Mean	Median	SD	2.5%	97.5%
PIN	0.295	0.292	0.083	0.146	0.460
α	0.231	0.221	0.126	0.041	0.516
δ	0.405	0.197	0.372	0.022	0.959
μ	474.3	136.2	2483.1	18.9	2614.7
ε_b	120.0	30.7	877.1	4.8	636.8
ε_s	72.7	25.0	436.4	3.3	356.4

The large gap between mean and median for μ , ε_b , and ε_s reflects the heavy right skew of these rate parameters: most markets have moderate trading activity, but a few high-volume markets pull the mean far above the median. The buy-sell asymmetry ($\varepsilon_b > \varepsilon_s$) is already visible here and becomes central to the calibration analysis in Section 6.

5.1.1 Bayesian vs. MLE

The comparison between Bayesian (M1) and MLE estimation is stark for the news-direction parameter δ . MLE places $\hat{\delta}$ at exactly 0 or 1 in 84.4% of markets; the Bayesian approach produces zero boundary solutions. This occurs because $\hat{\delta}_{\text{MLE}} = n_{\text{good}}/n_{\text{event}}$: if five event days are all classified as bad news, MLE concludes $\delta = 0/5 = 0$, declaring good-news events *impossible* in this market. This locks the entire parameter vector into a constrained optimum, distorting PIN. Thin markets (<30 days) face this most acutely, with an 89.2% boundary rate.

The Bayesian alternative is straightforward: $\delta \mid \text{data} \sim \text{Beta}(1 + 0, 1 + 5)$, giving a posterior mean of 0.14 with CI [0.01, 0.47] — “good news events are probably rare, but not impossible.” The other parameters do not face the same problem (α boundary rate: 0.1%). The issue is specific to δ because it conditions on event days, which are themselves a subset of an already short time series.

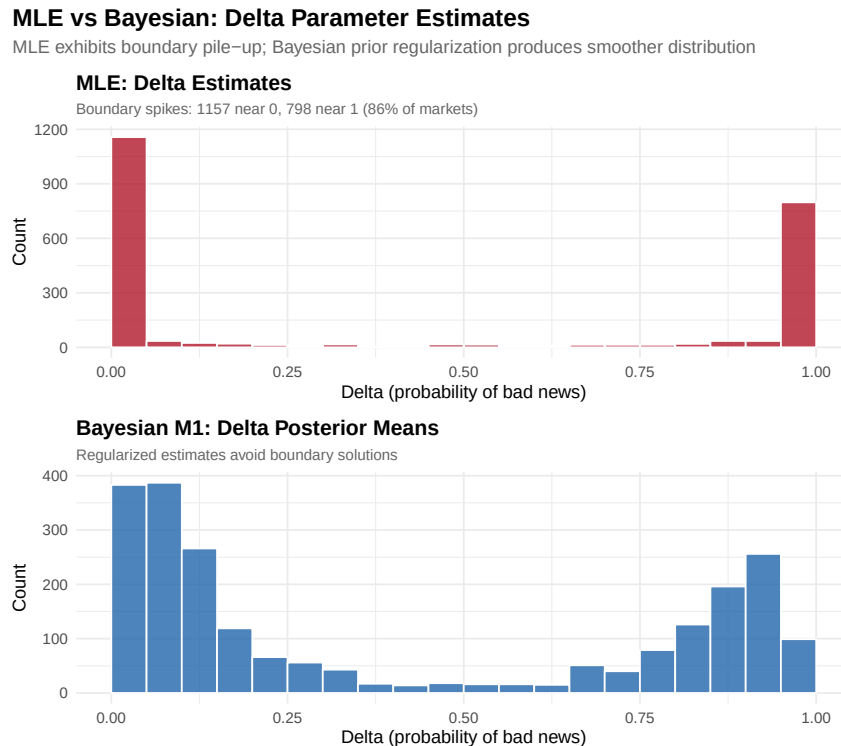


Figure 2: Distribution of δ estimates under MLE (top) and Bayesian M1 (bottom). MLE produces boundary solutions at exactly 0 or 1 in 84.4% of markets. Bayesian posterior means are distributed across the full range with no boundary spikes.

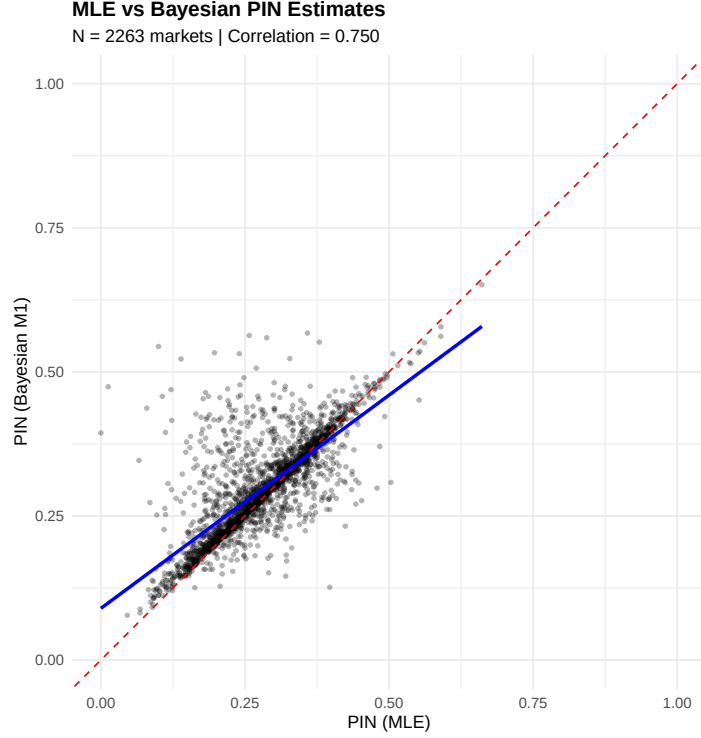


Figure 3: MLE vs. Bayesian PIN estimates across 2,263 markets ($r = 0.75$). The two approaches agree for moderate-PIN markets but diverge for extreme values, where MLE boundary solutions on δ distort the PIN calculation.

Median PIN is similar between the two approaches (MLE: 0.274, Bayesian: 0.292), and for well-traded markets the estimates are correlated ($r = 0.75$). The advantage of the Bayesian approach is not a different answer for data-rich markets, but an honest answer for data-poor ones: proper uncertainty quantification and avoidance of boundary artifacts.

5.2 M2: Hierarchical Partial Pooling

The MH steps required for market-level parameters reduce mixing efficiency relative to M1: median ESS is 1,575 (min 20). Figure 4 shows traceplots for the Politics category hyperparameters; both the category means (m_α , m_μ) and variances (s_α^2 , s_μ^2) mix well.

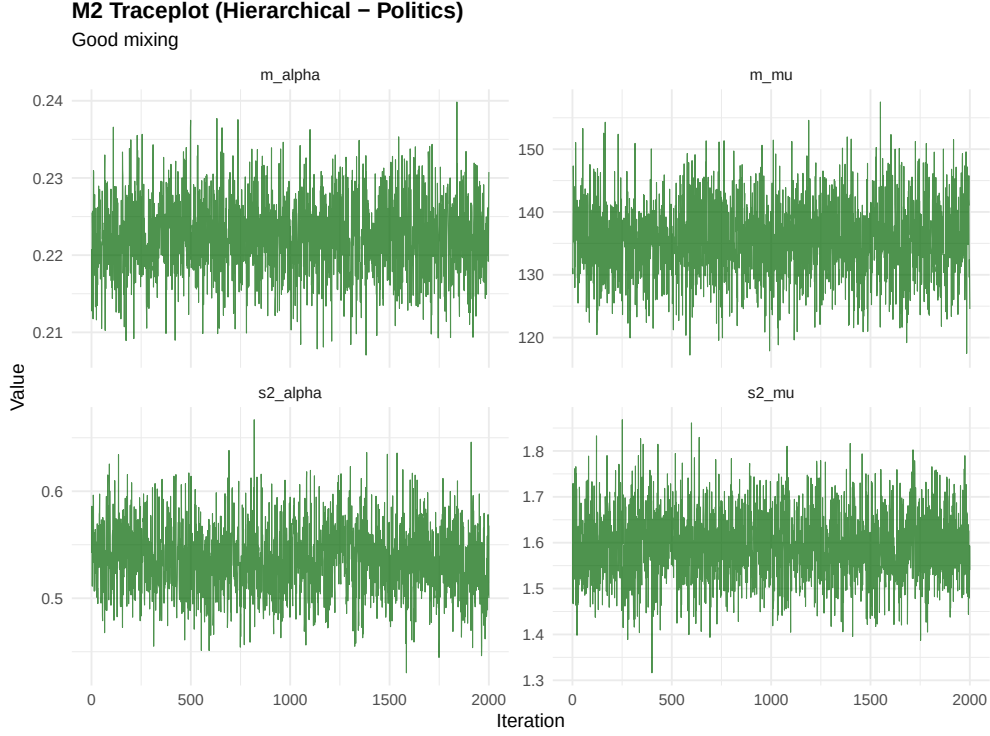


Figure 4: M2 traceplots for Politics category hyperparameters. Category-level means and variances mix well despite the MH steps required at the market level.

M2 produces median PIN of 0.301 with median 95% CI width of 0.183 — slightly higher PIN than M1 but with narrower uncertainty. Table 5 summarizes the posteriors across all markets.

Table 5: M2 posterior summary across 2,263 markets.

Parameter / Quantity	Mean	Median	SD	2.5%	97.5%
PIN	0.310	0.301	0.096	0.157	0.537
α	0.226	0.217	0.104	0.066	0.452
δ	0.476	0.373	0.352	0.038	0.957
μ	479.9	138.0	2478.4	19.8	2896.6
ε_b	98.1	28.7	589.0	4.9	533.1
ε_s	94.3	25.7	782.8	3.3	446.6

5.2.1 Prior Sensitivity

The three hyperprior specifications (tight, moderate, diffuse) on the category-level variance $s_{\theta,k}^2$ produce nearly identical PIN distributions. With 2,263 markets providing information to the hierarchy, the data overwhelms the hyperprior: the choice of hyperprior does not materially affect conclusions.

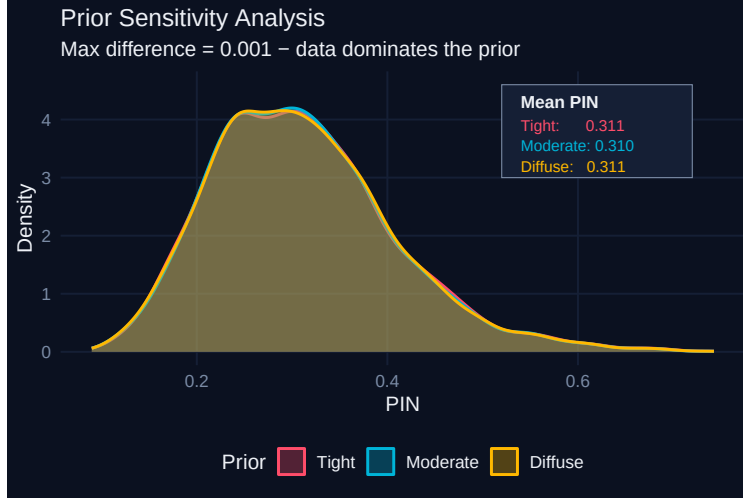


Figure 5: PIN distributions under three hyperprior specifications for M2. The three specifications produce nearly identical results, indicating the data overwhelms the prior.

5.2.2 Variance Regularization, Not Mean Shrinkage

The hierarchy narrows credible intervals for every market — 100% of markets have narrower CIs under M2 than M1. The effect is largest where data is scarcest: thin markets see 12.2% narrower CIs (liquid markets: 8.3%), and markets with fewer than 30 days see 15.2% narrower CIs (markets with >100 days: 5.2%).

However, only 47% of markets have their posterior mean pulled toward the category mean. The hierarchy’s primary contribution is narrowing uncertainty rather than shifting point estimates — variance regularization rather than classic mean shrinkage.

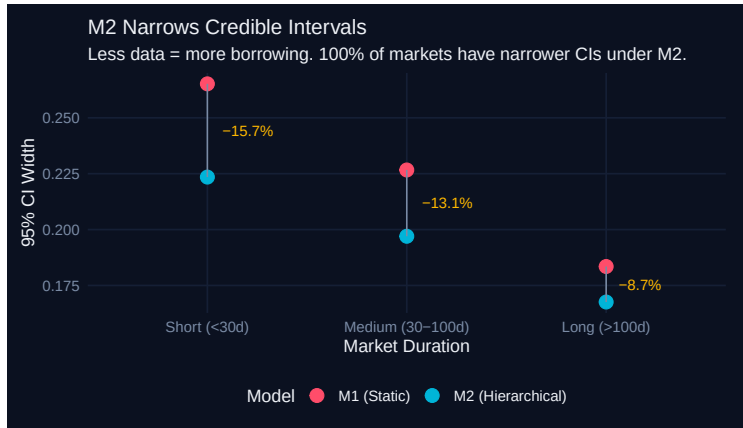


Figure 6: CI shrinkage from M1 to M2 as a function of market duration. Thin markets benefit most from hierarchical pooling.

Table 6 reports the learned category-level hyperparameters under the moderate prior. These represent the centers of the hierarchical prior distributions — the model’s learned expectation for a “typical” market in each category before incorporating that market’s own data. Sports has

the lowest prior event probability ($m_\alpha = 0.162$) but the highest prior informed trading intensity ($m_\mu = 251.3$), while Politics shows higher event frequency ($m_\alpha = 0.222$) with lower intensity ($m_\mu = 135.4$). The large variances on δ ($s^2 > 4$ across all categories) indicate substantial within-category heterogeneity in news direction, consistent with the MLE boundary problem: δ is the hardest parameter to pin down.

Table 6: M2 category-level hyperparameter estimates (moderate prior). Means (m) are on the natural scale; s^2 values are on the transformed scale (logit for α/δ , log for μ/ε). These are the learned hierarchical prior centers, not the market-level posterior averages reported in Section 5.3.2.

Category	α		δ		μ		ε_b		ε_s	
	m	s^2	m	s^2	m	s^2	m	s^2	m	s^2
Politics	0.222	0.54	0.457	5.74	135.4	1.59	32.5	1.45	29.3	1.58
Sports	0.162	0.58	0.461	7.50	251.3	1.73	35.5	1.71	31.2	1.95
Crypto & Fin.	0.219	0.43	0.484	5.05	151.5	1.72	39.9	1.33	38.6	1.39
Other	0.219	0.43	0.402	4.24	151.5	1.72	26.1	0.81	23.0	0.88

5.3 M3: Regime-Switching PIN

Despite the added complexity of FFBS for the regime sequence, mixing is strong: median ESS is 1,878, and 98.5% of markets have $\text{ESS} \geq 400$. Only 14 markets (0.6%) have ESS below 100. Figure 7 shows traceplots for a representative market; the regime-specific α values, transition probabilities, and overall PIN all mix well.

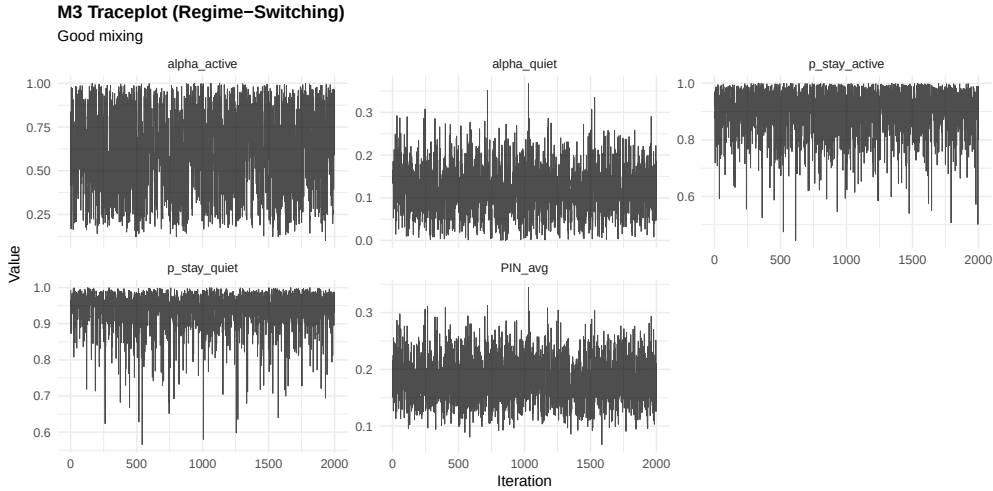


Figure 7: M3 traceplots for a representative market ($\text{ESS} = 1,878$). Regime-specific α values (α_{active} , α_{quiet}), transition probabilities ($p_{\text{stay active}}$, $p_{\text{stay quiet}}$), and overall PIN all show good mixing.

M3 produces the lowest median PIN at 0.222, with median 95% CI width of 0.177. The regime dynamics show that markets spend a mean of 28.2% of their life in the active regime, with highly

persistent states: the estimated probability of staying quiet is 0.938 and staying active is 0.893, consistent with the Beta(9,1) sticky-regime prior. Table 7 summarizes the posteriors, including regime-specific quantities.

Table 7: M3 posterior summary across 2,263 markets.

Parameter / Quantity	Mean	Median	SD	2.5%	97.5%
PIN (quiet)	0.125	0.109	0.073	0.030	0.316
PIN (active)	0.553	0.550	0.147	0.288	0.838
PIN (average)	0.228	0.222	0.070	0.116	0.393
α_{quiet}	0.083	0.062	0.067	0.011	0.248
α_{active}	0.665	0.670	0.169	0.355	0.944
δ	0.408	0.199	0.373	0.022	0.960
μ	471.8	136.1	2467.9	18.7	2614.9
ε_b	119.3	31.0	869.3	4.7	637.1
ε_s	72.4	24.9	433.9	3.3	359.4

5.3.1 Regime Separation

M3 finds strong evidence of two distinct information regimes. The quiet-regime α centers near 0.08, while the active-regime α centers near 0.67 — an approximately $8\times$ difference in information event rates. This two-regime structure appears in 99.4% of markets.

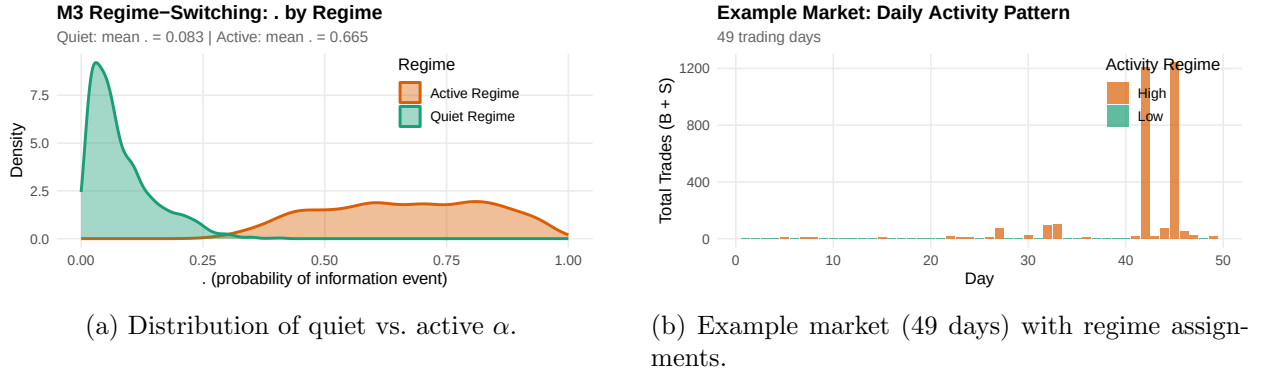


Figure 8: M3 regime separation. Left: two clearly separated distributions of α across regimes. Right: an example market showing volume spikes during inferred active periods (orange).

Markets spend 71.8% of their life in the quiet regime (average spell ~ 19 days) with short active bursts (~ 11 days). M1's single $\alpha = 0.22$ averages over both regimes, overstating quiet-period activity and understating active-period activity. M3's overall PIN is lower than M1 (0.222 vs. 0.292) not because there is less informed trading, but because M3 correctly weights the fact that markets are quiet most of the time.

5.3.2 Category Differences Dissolve Under Regimes

Under M2, categories appear to differ in PIN. Averaging individual market posteriors within each category,³ Sports shows the highest static PIN, driven by the least frequent information events ($\bar{\alpha} = 0.187$) but the largest bursts when they occur ($\bar{\mu} = 674$). This makes intuitive sense: in Sports, information events like injury reports are infrequent, but when they occur they can change outcomes drastically, so informed traders flood in. Politics shows the opposite pattern: information events are more frequent ($\bar{\alpha} = 0.243$) but each individual event spurs less intense informed trading ($\bar{\mu} = 488$).

Under M3, these category differences nearly vanish. Every category has similar quiet-regime PIN (~ 0.11) and similar active-regime PIN (~ 0.55). The categories differ in *when and how sharply* information arrives — Sports switches into active mode more sharply (4.1 switches, $\Delta\text{PIN} = 0.52$), while Politics drifts in and out more gradually — not in *how much* informed trading occurs.

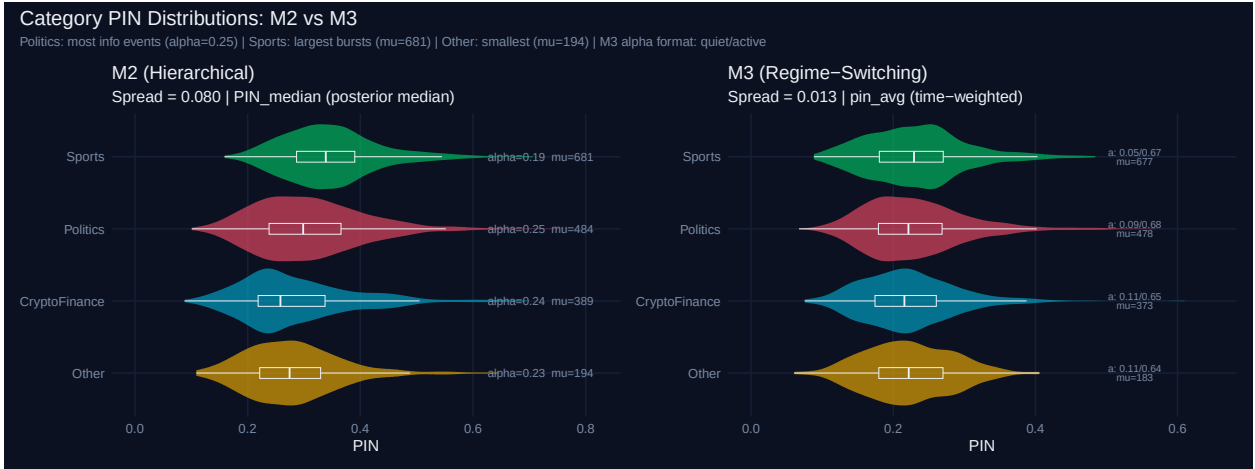


Figure 9: Category-level PIN under M2 (left) vs. M3 (right). M2 shows apparent category differences; M3 shows these disappear once regime structure is accounted for.

5.4 M4: Time-Varying PIN

M4’s sampling posed the greatest computational challenge. We initially attempted FFBS with a Laplace approximation to the Poisson mixture likelihood at each time step, but this produced a 9.8% acceptance rate — the Poisson mixture likelihood is not well-approximated by a Gaussian, so the Laplace proposals were poor. We replaced this with single-site MH (random walk on each $\text{logit}(\alpha_t)$ conditioned on its neighbors), which achieved a mean acceptance rate of 63.9% (median 61.6%, range 46–89%) — reasonable for random-walk MH, but at the cost of high autocorrelation between iterations.

The resulting median ESS across markets is just 16, with a worst case of 2. Figure 10 illustrates

³These market-level posterior averages differ from the hyperparameter means in Table 6, which represent the learned hierarchical prior centers on the transformed scale. The market-level averages incorporate both the prior and each market’s own data.

the problem: the PIN trace shows long, slow drifts rather than the rapid oscillation of a well-mixing chain, and σ_η^2 exhibits sudden jumps late in the run. For a market with $T = 115$ days, M4 must update 115 correlated $\alpha(t)$ values per iteration via single-site MH. Each update is accepted or rejected independently, so the chain can only make small local moves through the high-dimensional latent path.

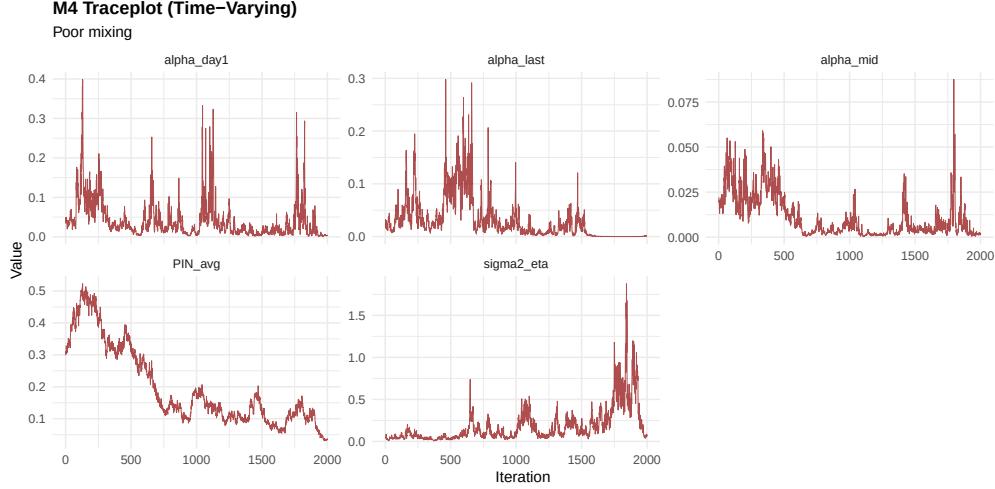


Figure 10: M4 traceplots showing poor mixing ($\text{ESS} = 2$, worst case). The PIN trace drifts slowly rather than oscillating, and σ_η^2 shows sudden jumps — hallmarks of a chain struggling in a high-dimensional correlated space.

Despite these limitations, M4 finds substantively similar dynamics to M3. The inferred $\alpha(t)$ path oscillates between approximately 0.15 and 0.90, with information arriving in waves. For example, in the “Trump declassifies JFK files in first 100 days” market (115 days), around days 55–80 a sustained burst of informed trading pushes the market price sharply upward toward the correct YES resolution — suggesting that informed trading can directly improve price accuracy in real time. M4’s smooth α path shows the same bursts as M3’s discrete regime switches, but with gradual buildup and decay rather than instantaneous transitions. The two models agree on *when* information arrives, providing mutual validation that information flow is not constant.

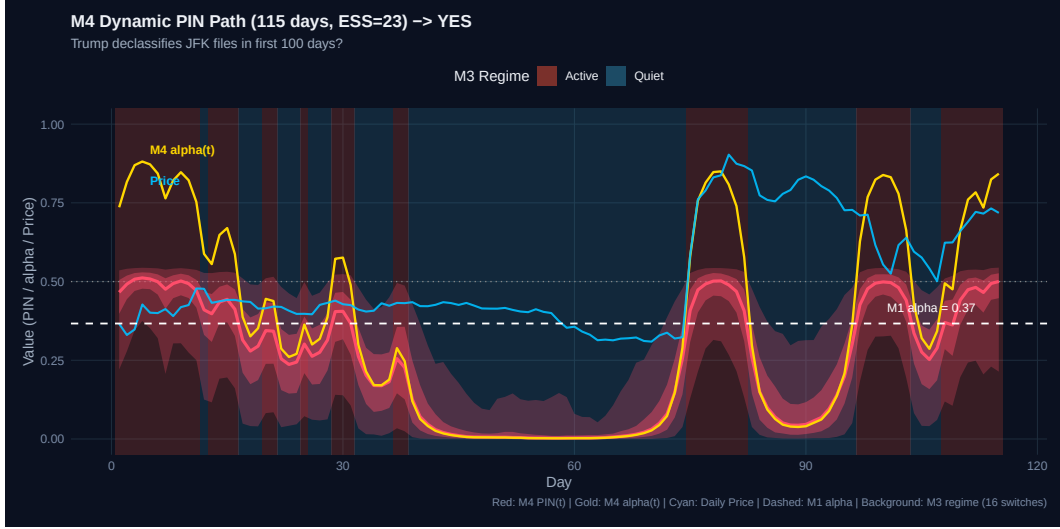


Figure 11: M4 time-varying α path for “Trump declassifies JFK files in first 100 days” (115 days). Gold line: M4’s continuous $\alpha(t)$. Dashed line: M1’s static $\alpha = 0.37$. Background shading: M3’s discrete regime assignments. ESS = 23.

M4 produces median PIN of 0.240 with median 95% CI width of 0.162, though the narrow CIs are an artifact of the chain underexploring the posterior rather than genuine precision. All M4 results should be interpreted with caution given the ESS limitation. A block-updating or particle MCMC approach to sampling the $\alpha(t)$ path jointly may improve mixing, but was beyond the scope of this project. Table 8 summarizes the posteriors.

Table 8: M4 posterior summary across 2,263 markets. All results should be interpreted with caution given ESS = 16 (median).

Parameter / Quantity	Mean	Median	SD	2.5%	97.5%
PIN (time-averaged)	0.247	0.240	0.074	0.119	0.412
α (time-averaged)	0.225	0.214	0.127	0.037	0.509
δ	0.395	0.183	0.370	0.022	0.959
μ	472.9	136.2	2468.0	18.8	2615.0
ε_b	119.6	31.3	869.0	4.8	636.8
ε_s	72.1	24.5	434.3	3.3	359.4
σ_η^2	0.326	0.154	0.516	0.061	1.541

5.5 Cross-Model Comparison

Table 9 summarizes PIN estimates across all four specifications.

Table 9: PIN estimates across specifications.

Model	Median PIN	Median 95% CI Width
M1 (static)	0.292	0.202
M2 (hierarchical)	0.301	0.183
M3 (regime-switching)	0.222	0.177
M4 (time-varying)*	0.240	0.162*

*M4 median ESS = 16; narrow CI reflects chain underexploration, not genuine precision.

Table 10 shows the pairwise correlations between models' PIN estimates. M1 and M2 agree most closely (0.854), which makes sense — they share the same structural assumption (constant α) and differ only in estimation strategy. M3 and M4 correlate most strongly with each other (0.838), as both allow α to vary over time. M2 and M3 correlate least (0.598): the hierarchy pulls estimates in a different direction than regime decomposition.

Table 10: Pairwise correlations of median PIN estimates across specifications.

	M1	M2	M3	M4
M1	1.000	0.854	0.710	0.826
M2		1.000	0.598	0.715
M3			1.000	0.838
M4				1.000

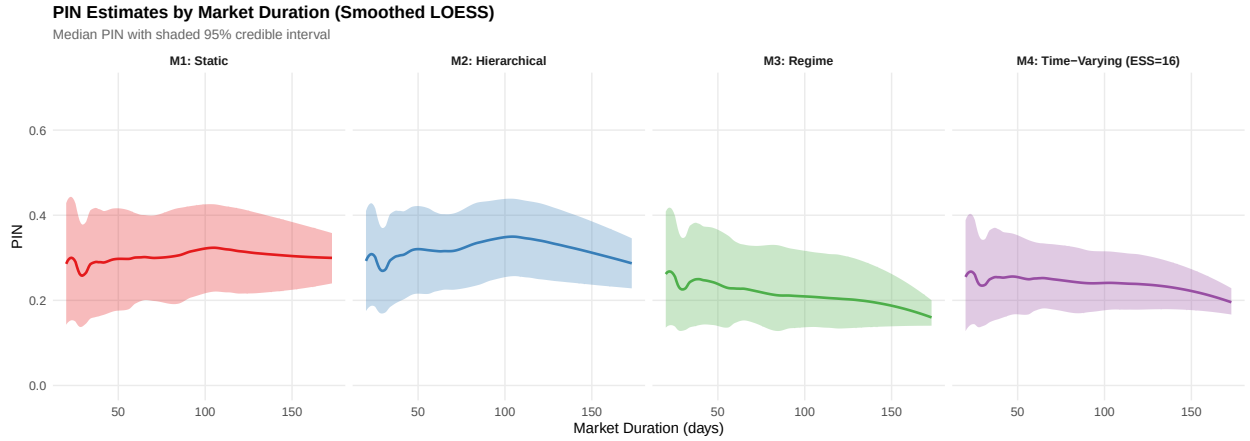


Figure 12: PIN estimates over time by specification. M1/M2 are flat (constant α). M3/M4 trend downward, reflecting that longer markets spend more time in quiet periods. Credible intervals narrow over time across all models as more data accumulates.

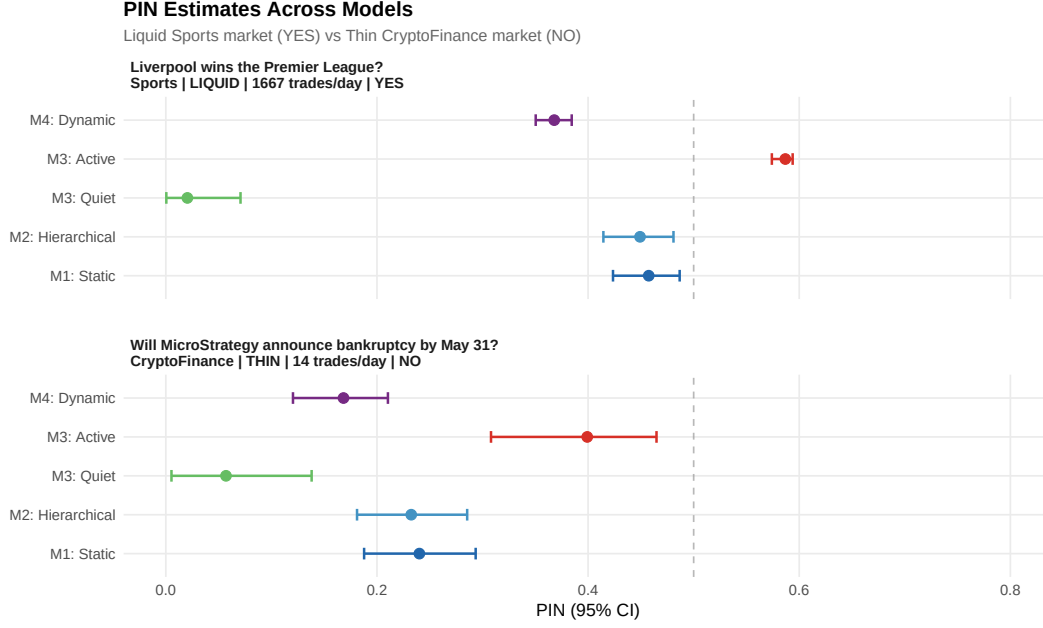


Figure 13: Forest plot comparing PIN posteriors across models for selected markets. Liquid markets show tight agreement; thin markets show wider divergence.

Table 11 compares model behavior on thin markets (30–50 average daily trades, $n = 427$) versus liquid markets ($n = 1,062$). M3’s PIN is nearly identical across thin and liquid markets (0.222 vs. 0.224), while M1’s PIN shifts substantially (0.268 vs. 0.311). This suggests M3’s regime decomposition is robust to market thickness, whereas M1’s static estimate is more sensitive to the volume of data. All models produce wider CIs on thin markets, as expected. Models diverge slightly more on liquid markets (cross-model SD of 0.040 vs. 0.033 for thin), likely because more data allows each specification to find its own distinct signal rather than all being pulled toward similar wide-prior estimates.

Table 11: Model comparison on thin ($n = 427$) vs. liquid ($n = 1,062$) markets.

Model	Median PIN		Median CI Width	
	Thin	Liquid	Thin	Liquid
M1	0.268	0.311	0.244	0.178
M2	0.286	0.310	0.217	0.163
M3	0.222	0.224	0.219	0.143
M4	0.232	0.243	0.196	0.136

6 PIN and Price Accuracy

6.1 Calibration by PIN Tercile

We stratify markets into PIN terciles (low, medium, high) and assess price calibration using the Brier score: $BS = \frac{1}{N} \sum (p_j - \omega_j)^2$, where p_j is the median market price and $\omega_j \in \{0, 1\}$ is the resolution outcome. A Brier score of 0 means perfect calibration (prices exactly match realized frequencies); 0.25 corresponds to an uninformative coin flip.

Table 12: Brier Score by PIN tercile (lower = better calibration). p -values from Wilcoxon rank-sum test on daily squared prediction errors between low and high PIN terciles ($\sim 40,000$ – $50,000$ observations per tercile).

Model	Low PIN	High PIN	p -value
M1	0.075	0.060	$< 10^{-40}$
M2	0.082	0.057	$< 10^{-101}$
M3	0.076	0.049	$< 10^{-136}$
M4	0.075	0.050	$< 10^{-120}$

All four models agree: markets with more informed trading produce better-calibrated prices. High-PIN markets consistently achieve lower Brier scores (0.049–0.060) than low-PIN markets (0.075–0.082), and the differences are highly significant ($p < 10^{-40}$ across all models). Table 13 reports the mean calibration error (average absolute deviation of the calibration curve from the 45° diagonal) by PIN tercile.

Table 13: Mean calibration error by PIN tercile (lower = better). M3 and M4 show monotonically decreasing error with PIN; M1 and M2 do not.

Model	Low PIN	Medium PIN	High PIN
M1	0.040	0.012	0.024
M2	0.034	0.014	0.021
M3	0.018	0.019	0.008
M4	0.026	0.016	0.014

Two patterns emerge. First, low-PIN markets are the worst calibrated across all models — this is unambiguous. Second, the relationship between medium and high PIN differs by model type: M3 and M4 (dynamic models) show a clear monotonic pattern where high PIN produces the best calibration, while M1 and M2 (static models) show medium PIN calibrating best. This suggests that dynamic models produce PIN estimates where the PIN–calibration relationship is cleaner.

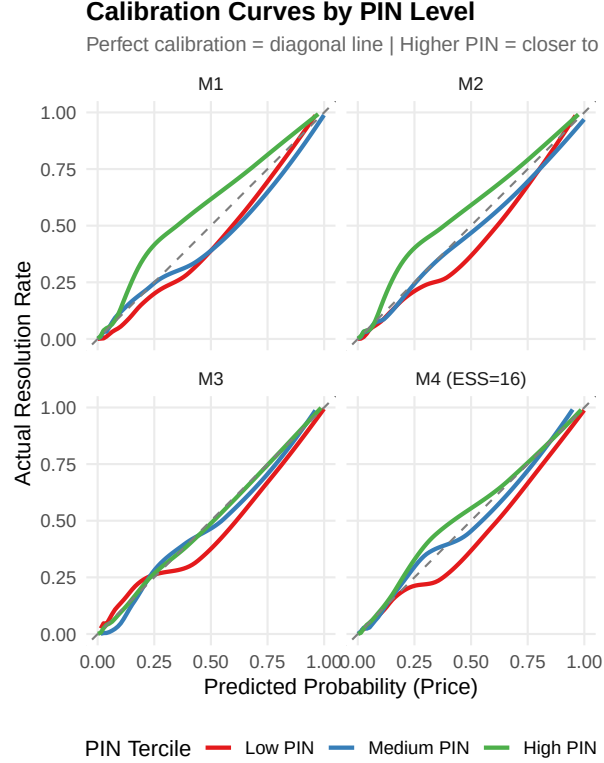


Figure 14: Calibration curves by PIN tercile. High-PIN markets (green) track the 45° diagonal most closely. M3 (bottom left) shows the tightest fit overall.

6.2 Which Model’s PIN Best Predicts Accuracy?

Having established that PIN predicts calibration, a natural follow-up is: which model produces the most useful PIN? This is not straightforward, because we do not observe the true fraction of informed trading in any market — PIN is latent. We therefore evaluate models from two complementary angles.

First, we compute the Widely Applicable Information Criterion (WAIC), which measures how well each model predicts the daily buy/sell counts it was trained on, with a penalty for model complexity. Lower WAIC indicates better in-sample predictive performance.

Second, we fit a logistic regression predicting whether a market’s median price correctly predicts the resolution outcome, using PIN as the predictor. A positive coefficient β_1 means higher PIN predicts more accurate markets; a larger β_1 means PIN is a more powerful discriminator. Unlike WAIC, which asks “who fits the trading data best?”, the logistic regression asks “whose PIN is most useful for predicting accuracy?”

Table 14: Model evaluation: WAIC (fit to trading data) and logistic regression (PIN predicting accuracy).

Model	WAIC ($\times 10^6$)	Logistic β_1	Logistic p
M1 (static)	12.20	0.58	0.527
M3 (regime)	12.95	2.88	0.013
M4	13.30	2.56	0.017
M2	28.18	1.23	0.130

M2 hierarchical p_{WAIC} unstable; not directly comparable.

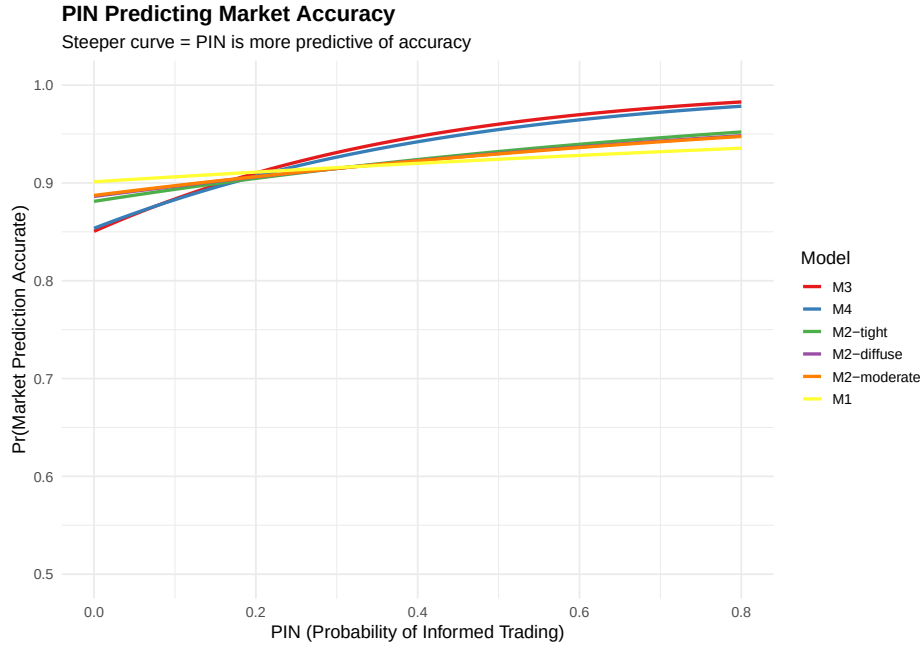


Figure 15: Logistic regression curves: probability of accurate price prediction as a function of PIN. M3 (steepest curve) produces the most discriminating PIN.

6.3 Fit vs. Usefulness

M1 wins on WAIC — it fits the observed trading data best, because M3’s regimes improve fit but not enough to overcome the complexity penalty on these relatively short time series (median 57 days). With longer-lived markets, M3’s regime structure would be better identified and the complexity penalty less dominant. However, M3 wins on the logistic regression — its PIN is the only one that significantly predicts price accuracy ($p = 0.013$), while M1’s PIN is not significant ($p = 0.527$).

This tension is the central finding of the model comparison. M1 fits *trading data* best, but M3 produces the most *useful* PIN. By separating quiet from active periods, M3 sacrifices some in-sample fit but creates PIN estimates that meaningfully discriminate between well-calibrated and

poorly-calibrated markets. The static models' PIN averages over regime structure, washing out the signal that makes PIN informative.

6.4 Uninformed Optimism and Why Informed Trading Helps

The mechanism connecting PIN to calibration is specific to Polymarket's structure. Uninformed traders on Polymarket bet YES 65% more than NO ($\varepsilon_b/\varepsilon_s = 1.65$), creating a chronic buy-side bias. But 83% of events in our sample resolve NO. This means uninformed traders are systematically optimistic — they push prices above the true probability.

The optimism bias varies substantially by category: Sports markets exhibit the strongest bias ($\varepsilon_b/\varepsilon_s = 2.10$), followed by Politics (1.72), Other (1.39), and Crypto & Finance (1.08). Crypto & Finance markets are nearly balanced, suggesting that traders in these categories may be more sophisticated or less prone to directional bias than sports bettors.

Informed traders exploit this optimism by betting against it, which pushes prices back toward truth. In markets with high PIN, this correction is stronger, and the resulting prices are better calibrated. In markets with low PIN, the uninformed optimism persists, and prices overstate the probability of YES outcomes. Consistent with this, high-PIN markets spend more time in M3's active regime (34.2% vs. 21.1% for low PIN), providing more opportunity for informed correction. High-PIN markets also exhibit lower price volatility (mean price SD of 0.044 vs. 0.075 for low PIN), suggesting that informed trading resolves uncertainty rather than creating it.

This explains why PIN predicts accuracy in prediction markets: informed trading acts as a corrective force against the structural bias of the uninformed majority.

7 Discussion

7.1 PIN as Cost vs. Quality

In equity markets, [Easley and O'Hara \(2004\)](#) show that informed trading imposes a cost: uninformed investors face adverse selection, demand higher returns, and may exit the market entirely. In prediction markets, the same mechanism operates but the implications are reversed. The purpose of a prediction market is to produce accurate probability forecasts, and when informed traders push prices toward truth, everyone who reads the price benefits. The uninformed traders who lose money are, in effect, subsidizing the production of public information.

Our results make this concrete. Uninformed traders exhibit a systematic optimism bias ($\varepsilon_b/\varepsilon_s$ ranging from 1.08 in Crypto & Finance to 2.10 in Sports), and informed traders correct this by betting against it. The high-PIN markets where this correction is strongest produce the best-calibrated prices. This creates a tension for platform design and regulation: reducing informed trading would protect uninformed traders but degrade forecast accuracy. For regulators debating whether to restrict prediction markets (as the CFTC has for U.S. election markets), the consumer protection case must be weighed against the public information value.

One complication is that the PIN model cannot distinguish genuine informed trading from market manipulation — both appear as excess directional activity. This is particularly concerning for election markets, where the relationship may be self-reinforcing: a large trader pushing prices toward a candidate could shift public perception, influence voter behavior, and cause the outcome to match the manipulated price. Our PIN-calibration finding cannot distinguish “informed traders make prices accurate” from “large traders make prices self-fulfilling.” Separating these mechanisms remains an open challenge, especially as prediction market prices receive increasing media attention during elections.

7.2 Robustness and Limitations

Robustness. Our primary order flow specification combines YES and NO token activity ($B_{jd} = \text{buy-YES} + \text{sell-NO}$, $S_{jd} = \text{sell-YES} + \text{buy-NO}$). We initially tested a specification using only YES token trades (buy-YES vs. sell-YES), which ignores information contained in NO token activity. Switching to the combined specification reduced PIN by approximately 4–5 percentage points and δ dropped substantially, but α was unchanged and M3’s regime structure remained robust, confirming the core findings are not artifacts of the order flow definition. We also tested whether δ correctly predicts the direction of resolution: if posterior mean $\delta > 0.5$ (more good-news events than bad-news events), we predict the market resolves YES; if $\delta < 0.5$, we predict NO. M1 and M3 achieve 58% directional accuracy, which is better than a 50% coin flip but below the 83% accuracy of simply predicting NO for every market — a strong baseline given that 83% of markets in our sample resolve NO.

Limitations and future work. M4’s median ESS of 16 means its posterior is not fully explored; all M4 results should be interpreted with caution. Improving M4’s sampler — via block updates, particle MCMC, or Hamiltonian methods — would determine whether continuous dynamics add value beyond what M3’s discrete regimes capture. WAIC comparison across models is imperfect: M2’s hierarchical p_{WAIC} is unstable, and M4’s log-likelihood is conditional on the α path, making direct comparison with M1 and M3 problematic. Our sample has a median market duration of only 57 days, which limits M3’s ability to detect regime structure — many markets barely have time for one full regime switch. Applying this framework to longer-lived markets (e.g., presidential election markets running 6–12 months, or perpetual cryptocurrency markets) would give M3 more data to identify regimes and could shift the WAIC comparison in its favor. The PIN model assumes informed traders arrive via independent Poisson processes, which may not hold if informed traders strategically time their trades or if multiple informed traders coordinate. More fundamentally, the model treats all non-noise directional trading as “informed,” but cannot distinguish genuine private information from market manipulation — both produce the same signature of excess directional activity. Finally, running M3 on live trading data could provide a real-time signal of forecast reliability: a market entering an active regime with high PIN may be producing more trustworthy prices than one sitting in a quiet regime dominated by uninformed optimism.

8 Conclusion

We estimated the Probability of Informed Trading on 2,263 resolved Polymarket markets using four Bayesian specifications. Three findings stand out.

First, Bayesian estimation solves a real problem. MLE produces boundary solutions for δ in 84.4% of markets; the Bayesian approach produces zero. This is not a theoretical concern — boundary δ locks the parameter vector into constrained optima that distort PIN. The Bayesian advantage is specific to thin markets: for well-traded markets with long histories, Bayesian and MLE estimates converge. The hierarchical specification (M2) adds further value by reducing credible interval widths by 15.2% for markets with <30 days, without forcing estimates toward the category mean.

Second, model structure matters for what you can learn from PIN, even when it doesn't win on fit. M1 fits trading data best (lowest WAIC), but M3's regime-switching PIN is the only specification that significantly predicts price accuracy ($\beta = 2.88$, $p = 0.013$). The act of separating quiet from active periods creates PIN estimates that carry information about forecast quality.

Third, informed trading improves prediction market prices. High-PIN markets are consistently better calibrated (Brier scores 0.049–0.060 vs. 0.075–0.082 for low PIN). The mechanism is specific: uninformed traders are net optimistic ($\varepsilon_b/\varepsilon_s = 1.65$) in a sample where 83% of events resolve NO, and informed traders profit by correcting this bias. Unlike equity markets where PIN represents adverse selection cost, in prediction markets PIN signals forecast quality.

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